MILITARY SPECIFICATION

PROJECTILE, 155MM: HE, M692/731
METAL PARTS ASSEMBLY

This amendment forms a part of Military Specification MIL-P-63065A (AR), dated 18 Jan 1982, and is approved for use by the U. S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1.1, Standards, Military, MIL-STD-410: Delete title of MIL-STD-410 and substitute "Nondestructive Testing Personnel Qualification and Certification."

PAGE 3

2.2, Other publications: Add the following:

"AMS Specification Control of Forgings Requiring First Article Approval.

(Application for copies should be addressed to Society of Automotive Engineers, 400 Commonwealth Drive, Warrandale, PA 15906)."

PAGE 5

3.7.3: Delete this paragraph in its entirety.

Add the following new paragraphs 3.7.4 and 3.7.5:

"3.7.4 Monitoring of hot forge temperature. When bases and ogives are hot forged, the forger shall provide continuous, automatic monitoring of slug temperature prior to hot forging. Details shall be provided in accordance with paragraph 3.8.1.

3.7.5 Storage. Forgings, semi finished, or finished components shall be stored indoors. Forgings shall be processed into finished components in a sequential forging lot basis; i.e., first in first out."

FSC 1320

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3.9.1: Delete first three sentences and replace with: "Dimensions marked tool control shall be gaged during First Article Inspection and after eight hours maximum production on each machine in operation and prior to any tooling change."

3.9.2: Add the following to the end of the paragraph: "Dimensions to be gaged for process control are to be assured to a .40% AQL."

3.8.1: Add new paragraph 3.8.1 as follows:

"3.8.1 First article approval of aluminum forgings. First article approval of forgings, forging drawings, process and controls to be used in producing subsequent production forgings shall be obtained from the government technical agency prior to initating production. The procedures for obtaining first article approval shall be in accordance with AMS 2375."

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Table I - First article inspection: Add the following:

Ogive - "Forging	*	-	3.8.1	*"
Base - "Forging	*	-	3.8.1	*"

Notes - "*See appropriate requirement paragraph".

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Table I, First article inspection: Change "Projectile metal parts (painted)" to "Projectile metal parts".

PAGE 10

4.4.1.1.1: Add the following to the end of this paragraph:

"The maximum lot size shall be 2000 bodies. If a mill heat exceeds 2000 bodies, the mill heat shall be divided into any convenient number of heat treatment lots that would preclude the need for forming miscellaneous heat treatment lots."

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4.4.2.1: Major 111, second line: Change "undercoat" to "undercut".

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4.4.2.1:

Major 118: Delete "base" and substitute "body".

Major 119: Change "(2)" to "(2 places)". Change "body shoulder" to "body thread pitch diameter, body shoulder".

Minor 205: Delete in its entirety.

Minor 211: Change "3.1" to "3.11".

Add Note "Major 118, 119 - These inspections are not required when components are to be supplied as assemblies."

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4.4.2.3, Major 114: Change "3.11" to "3.1".

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4.4.2.3:

Minor 209: Change "3.5" to "3.1"

Minor 212: Change "3.11" to "3.1".

Minor 213: Change "3.1" to "3.11".

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4.4.2.4, Minor 202: Change "3.1" to "3.11".

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4.4.2.6:

Major 150: Delete "of ogive and thread" and substitute: "projections (2 places) of ogive thread".

Add Note: "Major 150 - This inspection is not required when components are to be supplied as assemblies."

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4.4.2.6:

Minor 207: Change "3.11" to "3.1".

Minor 212: Change "gage" to "visual".

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4.4.2.7:

Major 101: Change "100%" to ".40%".

Minor 201: Delete in its entirety.

Minor 204: Change "3.11" to "3.1".

Minor 205: Change "3.1" to "3.11".

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4.4.2.8, Major 102: Delete: ".40% AQL" and substitute: "100%".

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4.4.2.9, Minor 205: Change "3.1" to "3.11".

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4.4.4.2: Add the following new paragraph:

"4.4.4.2 Certification of nondestructive test personnel. All personnel operating or calibrating nondestructiver test equipment shall be qualified and certified in accordance with MIL-STD-410."

4.5.1.1, first line: Insert the following prior to the first sentence:

"The surface of each body in the area to be tested shall be prepared by machining or grinding a flat of sufficient depth, .030 inches minimum, to remove scale and get below the decarburization zone."

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4.5.1.2, line 7: Delete last sentence and replace with the following:

"The test shall be made in accordance ASTM Method E-103 and the following:

a. The surface of the body in the area to be hardness tested shall be prepared by machining or grinding a flat of sufficient depth, .015 inches minimum, to remove scale and get below the decarburization zone. (Note: Flats and impressions shall be completely removed by final machining.)

b. Hardness tester periodic verification shall be performed using brinnel test standards manufactured in accordance with ASTM Method E-10 in lieu of production samples.

c. All hardness test measurements shall be automatically recorded.

d. The hardness tester shall use a standard 10MM brinnel ball. The hardness tester shall determine hardness in accordance with the following test sequence. First, the tester shall clamp the body to be tested with a 10,000-500 lb. clamping load. Second, apply a minor load of approximately 10% of the 3000 kg of total test load. Third, apply the total test load. The clamping load will not be exceeded when the test load is applied. Fourth, relax the total test load back to the minor load. Fifth, automatically measure this penetratoir travel from initial minor load position to new minor load position following application of the total test load. Sixth, display hardness and release the body."

4.5.1.4: Delete in its entirety and substitute with the following:

"4.5.1.4 Yield strength and percent elongation of body. The hardest and two (2) softest bodies from each standard heat treatment lot or the two (2) hardest and four (4) softest bodies from each miscellaneous heat treatment lot as determined by the hardness test in 4.5.1.2 shall be subjected to this test. Sample for mechanical property testing shall be selected after heat treatment of the entire heat treatment lot. Two tension test specimens conforming to the largest obtainable round type specimen as prescribed by ASTM Method E-8 shall be taken

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longitudinally from opposite sides of each body, midwall of the band seat area, and two specimens as prescribed by ASTM Method E-8 shall be taken longitudinally from opposite sides of each body, midwall of the area of the body one inch from the front end. The number of specimens and their location are for the convenience of the contractor and government and shall not restrict the contractor or government from testing specimens at other locations to determine that the mechanical properties of the body meet the drawing requirements. The criteria for performing the tests shall be as prescribed by ASTM Method E-8. Failure of any test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot. At the option of the contractor the entire lot may be reheat treated and retested; or a new set of bodies may be selected and tested from a segment of the heat treat lot having a tighter hardness range than the heat treatment lot as originally sampled. If an acceptable hardness range can be established for this heat treatment lot, bodies falling within this range will be accepted. All other bodies will be rejected. Rejected bodies may be reheat treated and retested."

4.5.1.4.1: Delete in its entirety.

PAGE 29

4.5.3.1, Title: Change "Band seat" to "Magnetic particle inspection of band seat".

First sentence: Add the following to end of sentence: "to comply with the visual standards of 3.5.2".

4.5.1.5, line 13: Delete the following: "except the elongation shall be measured to the nearest thousandth prior to conversion into percent elongation. Percent elongation shall be recorded to the nearest tenth percent."

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4.5.4: Delete first sentence of second paragraph.

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4.5.8.1, line 7: Change "eight hours of" to "twenty four hours of continuous".

4.5.9: Add the following between the second and third sentence:

"After 10,000 fiberglass wrapped bodies have been produced and accepted, a reduced sample of one body from each 24 hours of continuous production or fraction thereof from each spindle in operation shall be subjected to this test to determine compliance with 3.4."

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6.4: Add "3.8.1" to the first paragraph for Procedure Report.

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6.13: Add the following paragraph:

"6.13 To minimize the need for reheat treating material or for having miscellaneous heat treatment lots it is recommended that the contractor process a preproduction sample to establish what should be an acceptable hardness range for each mill heat. This hardness range could then be used for screening bodies during processing of the heat treatment lot from that mill heat. When using a hardness range acceptance will still be in accordance with para. 4.5.1.4."

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